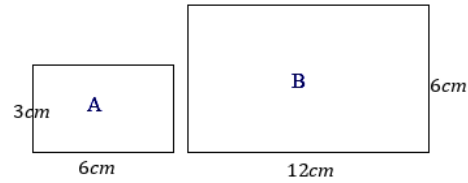


Perimeters and areas of similar figures

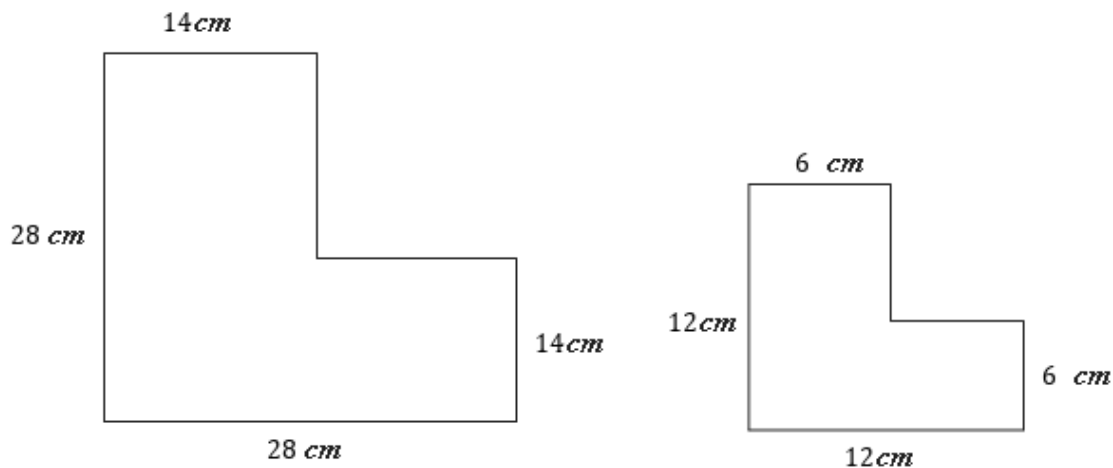


1 Consider the two similar rectangles shown.

- a Find the area of Rectangle A.
- b Find the area of Rectangle B.
- c What is the ratio of the area of Rectangle A to Rectangle B?
- d Is it true in this case that if the matching sides of two similar figures are in the ratio $m : n$, then their areas are in the ratio $m^2 : n^2$?

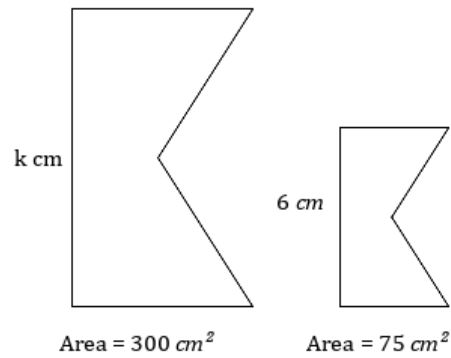


2 The corresponding sides of two similar figures are in the ratio 7 : 3.



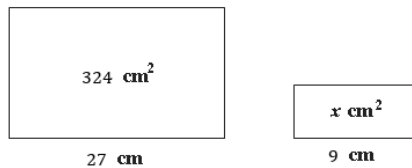
- a State the length scale factor from the left to the right figure.
- b State the area scale factor from the left to the right figure.

- 3 The figures in each pair are similar.
Find the value of k .

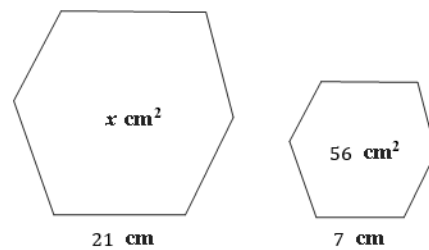


- 4 Find the value of x for each of the following figures:

a

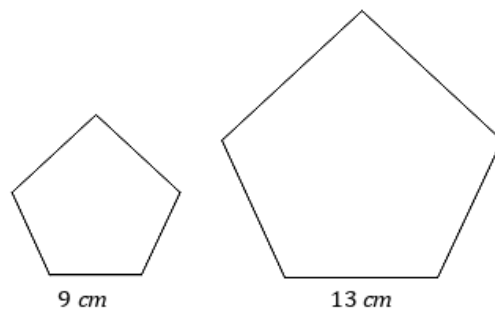


b



- 5 Two similar pentagons have sides of length 9 cm and 13 cm.

- a Find the ratio between the smaller area and the larger area.
- b If the area of the smaller pentagon is 81 cm^2 , find the area of the larger pentagon.



- 6 A triangle has side lengths of 7 cm, 10 cm and 16 cm. A similar triangle to it has 9 times the area of the first triangle.

- a Find the length scale factor between the two triangles.
- b Find the side lengths of the second triangle.
- c State the relationship between the perimeters of the two triangles.

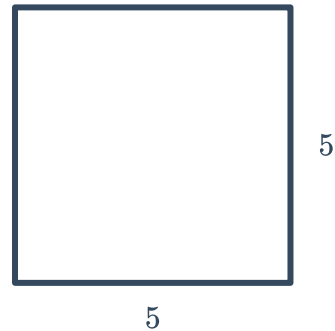


- 7 A square of side length 6 cm is enlarged using the scale factor 3. Find the area of the enlarged square.
- 8 Two similar parallelograms have matching sides whose lengths are in the ratio 7 : 2. Find the ratio of their areas.
- 9 The diameter of a circle is tripled. What happens to its area?
- 10 Consider two similar parallelograms with a matching sides in the ratio 6 : 8.
- a Calculate the area of the larger parallelogram, if the area of the smaller parallelogram is 72 cm^2 .
 - b Find the length of the base of the smaller parallelogram , if the length of the base of larger parallelogram is 12 cm.
- 11 The corresponding sides of two similar triangles are 8 cm and 40 cm. If the area of the smaller triangle is 24 cm^2 , find the area of the larger triangle.
- 12 The cost of constructing a garden bed is dependent on the area it covers. A rectangular garden bed measuring 16 ft by 24 ft costs a known amount to construct. If you decided to create a similar garden bed, measuring 20 ft by 30 ft, by what factor would the cost increase?
- 13 Consider two similar ceilings where the first has dimensions 5 m by 4 m, and the second has dimensions 20 m by 16 m.
- a Find the length scale factor.
 - b Find the area scale factor.
 - c If the smaller ceiling took 1.5 L of paint to cover it, determine the amount in liter of paint required to cover the second ceiling.
- 14 Consider the formula $A = \frac{1}{2}bh$. Determine the effect of each of the following have on A .
- a The values of b and h double.
 - b The value of b increases by a factor of 2 and the value of h decreases by a factor of 2.
- 15 Consider the formula $A = \pi r^2$. Determine the effect of each of the following have on A .
- a The value of r triples.
 - b The value of r decreases by a factor of 4.



Additional questions

- 16 Emily is knitting a rectangular blanket. So far the blanket measures $5 \text{ ft} \times 5 \text{ ft}$. If Emily increases both the length and the width of the blanket by 1 ft, how many times bigger will the area of the finished blanket be?



- 17 A triangle with an area of $\frac{3}{5} \text{ m}^2$ is dilated by a factor of 10. Find the area of the new triangle.
- 18 A regular pentagon has been enlarged with a scale factor of 3.25. Describe the relationship between the perimeter of the two pentagons.